SUPPORTING ADULT GUIDANCE

Area of a Circle

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner derives and uses the area formula for a circle, $A = \pi r^2$, and discovers that the area is proportional to the square of the radius. The emphasis is on **building the formula** from a rearrangement, and on understanding why doubling the radius quadruples the area — not on memorising a rule.

Driving Question: *How much space is inside a circle — and how does it depend on the radius?*

The mathematics, in plain terms

The **area** is the flat space inside a circle, in square units. If you cut a circle into many thin **sectors** (pizza slices, each bounded by a curved arc) and lay them alternately point-up and point-down, they form a shape that approaches a **parallelogram**. Its base is half the circumference, πr, and its **perpendicular height** is the radius, r. Since the area of a parallelogram is base × perpendicular height, the area of the circle is $\pi r \times r$, which is $\mathbf{A = \pi r^2}$. Because the area depends on r^2, it is proportional to the square of the radius: a graph of area against r^2 is a straight line through the origin with gradient π. And doubling the radius multiplies the area by $2^2 = 4$. To work backwards from area to radius, divide by π and take the square root.

Radius squared, not radius

The single most useful idea to protect is that the area depends on the radius *squared*. A learner who has watched the sectors become a parallelogram understands *why* $A = \pi r^2$, and is far less likely to muddle it with the circumference. Two habits help most: always **square the radius first, then multiply by** π (not the other way, and never π times r on its own); and, when a diameter is given, **halve it to the radius before squaring**. The doubling result — twice the radius, four times the area — is worth dwelling on, because it is surprising and it is the heart of what 'depends on r^2' means.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Circle to Parallelogram: rearranging the same sectors into a parallelogram of base πr and perpendicular height r, to build $A = \pi r^2$.	~15 min
Movement	The (r^2, A) Plotter: the point G traces a straight line as the circle changes, revealing the gradient π and the equation $A = \pi r^2$.	~8 min

Reflection	Judging true/false statements about area, centred on 'doubling the radius quadruples the area', and giving the reason.	~10 min
Creativity	Estimating the area of real objects (a coin, a plate, a pizza, a pond) with $A = \pi r^2$.	~5 min
Rest	A pause to notice which built the formula for you — the rearranged sectors, or the points on the graph.	~4 min
Nutrition	Contextual (Mode A): pizza — a 12-inch pizza has four times the area of a 6-inch one, because area depends on r^2.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using π r instead of π r^2

Confusing area with circumference. Repair: “Is this the distance around, or the space inside? For the space inside, what do we do to the radius first?”

Multiplying by π before squaring

Working $(\pi r)^2$ or π times r. Repair: “Square the radius first, then multiply by π. What is r^2 here?”

Using the diameter as the radius

Repair: “Is that measurement across the whole circle or only to the edge? If it is the diameter, what is the radius?”

Thinking doubling the radius doubles the area

Repair: “The area depends on r^2. If r doubles, what does r^2 do — double, or something more?”

Believing the rearranged pieces are triangles

Repair: “Look at the edge of each piece — is it a straight line or a curved arc? What shape has a curved edge like that?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: squaring a number, the area of a parallelogram, and the parts of a circle. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the reasoning — and the squaring — with the learner:

- “Is this asking for the distance around, or the space inside?”
- “What is the radius? Now, what is the radius squared (r^2)?”
- “Square the radius first, then multiply by π — what do you get?”
- “If the radius doubled, how many times bigger would the area be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise A = π r^2?”
Understanding *why* — the parallelogram of base π r and height r — matters more than memorising. If they can rebuild it from the rearrangement, that is the goal.

“They keep getting circumference and area mixed up.”
Very common. Anchor it: circumference is a *distance* (one r, C = π d), area is a *space* (r^2, A = π r^2). The square is the giveaway that it is area.

“Why make such a point of doubling the radius?”
Because 'depends on r^2' is abstract until they see it: twice the radius gives four times the area. It explains why a large pizza is such good value, and it is the idea the whole lesson rests on.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.